CAPSTONE PROJECT · BUSINESS ANALYTICS

Hotel Booking Analytics

Reducing cancellation losses and forecasting demand for a City Hotel (Lisbon) and a Resort Hotel (Algarve)

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Tools: Python · Excel · Power BI



PROBLEM STATEMENT

More than 1 in 3 bookings is cancelled

CANCELLATION RATE

37.1%

44,198 of 119,208 bookings

BOOKED VALUE LOST

€16.7M

= 64% of revenue earned

REALISED REVENUE

€26.0M

Jul 2015 – Aug 2017

Management has no single view of where cancellations come from, which upcoming bookings are at risk, or how much demand to expect – so pricing, deposit and overbooking decisions are reactive.

Goal: build a business analytics system (Excel + Power BI) with a simple cancellation risk flag and a 12-month forecast so revenue managers can cut cancellation losses, price by season and segment, and plan capacity.

RESEARCH OBJECTIVES

Five research objectives

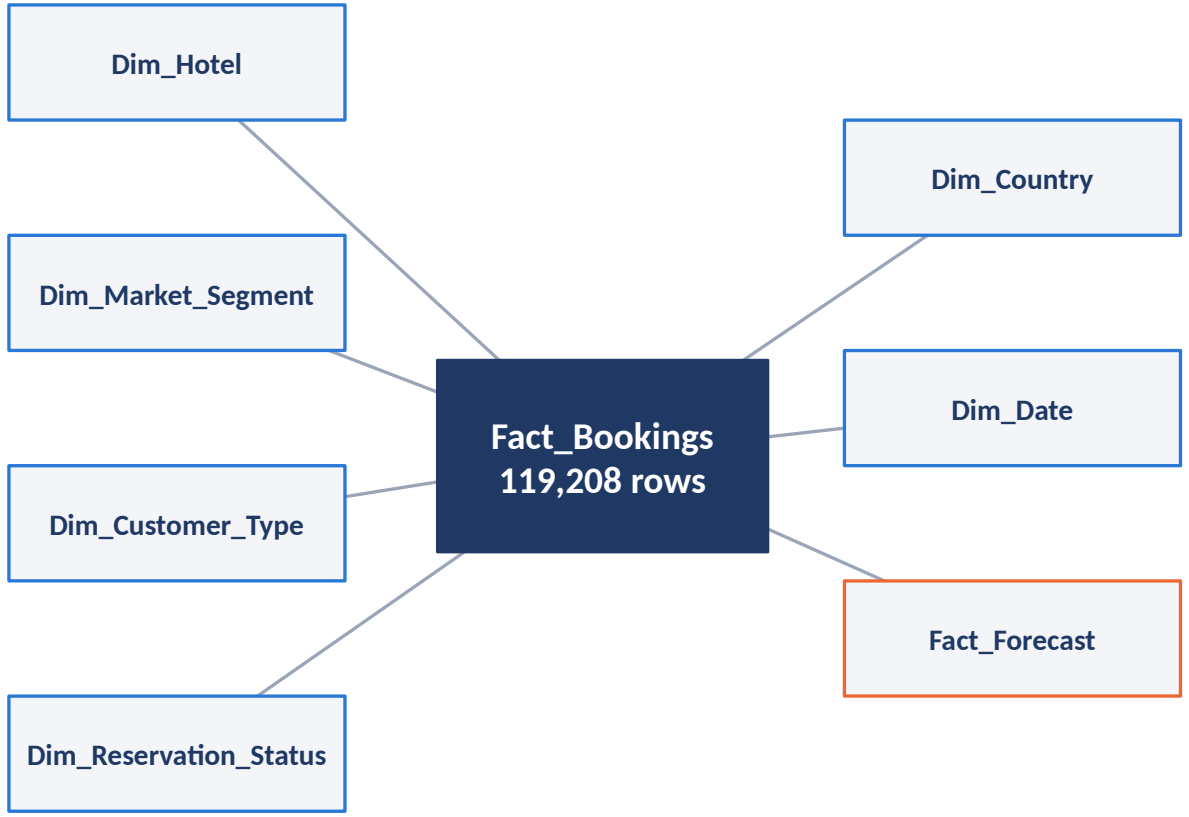
RO1	Measure overall performance and seasonal patterns – volume, revenue, ADR, length of stay
RO2	Identify the drivers of cancellation and quantify the revenue lost
RO3	Compare market segments and distribution channels on volume, value and reliability
RO4	Profile guests to find loyalty and engagement opportunities
RO5	Forecast monthly bookings and revenue for the next 12 months

Blue = understand the past Orange = look ahead · Business Analytics approach

DATA PREPROCESSING

Five tables cleaned into one star schema

1. **Load** 119,390 raw bookings + 4 lookup tables (hotel, segment, customer type, status)
2. **Fill gaps** children → 0, country → "Unknown", agent/company → "none" flags
3. **Remove** 182 invalid rows (0 guests, negative ADR, ADR of 5,400)
4. **Keep** 32,995 look-alike rows – no guest ID, so they are most likely rooms in the same group booking
5. **Enrich** 17 new fields: arrival & booking date, nights, guests, booking value, realised & lost revenue, lead-time band, season, risk flag



Lookups (blue) filter the fact table one-to-many; Fact_Forecast holds monthly actuals + forecast.

RO1 · OVERALL PERFORMANCE

Headline performance

BOOKINGS

119,208

2 hotels, 26 months

CANCELLATION RATE

37.1%

City 41.8% · Resort 27.8%

REALISED REVENUE

€26.0M

City 55% · Resort 45%

ADR (STAYED)

€100.17

City €106 · Resort €91

AVG STAY

3.39 nights

City 2.9 · Resort 4.1

REPEAT GUESTS

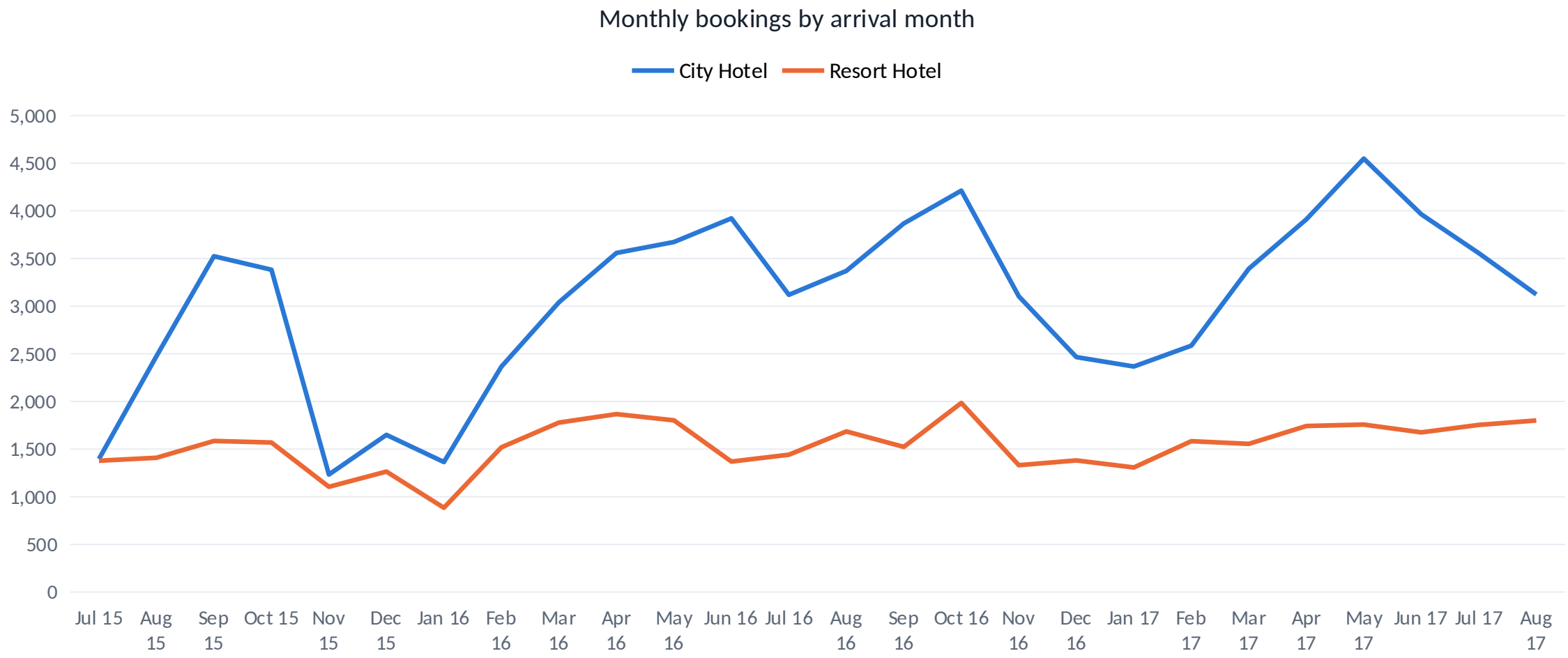
3.1%

loyalty is low

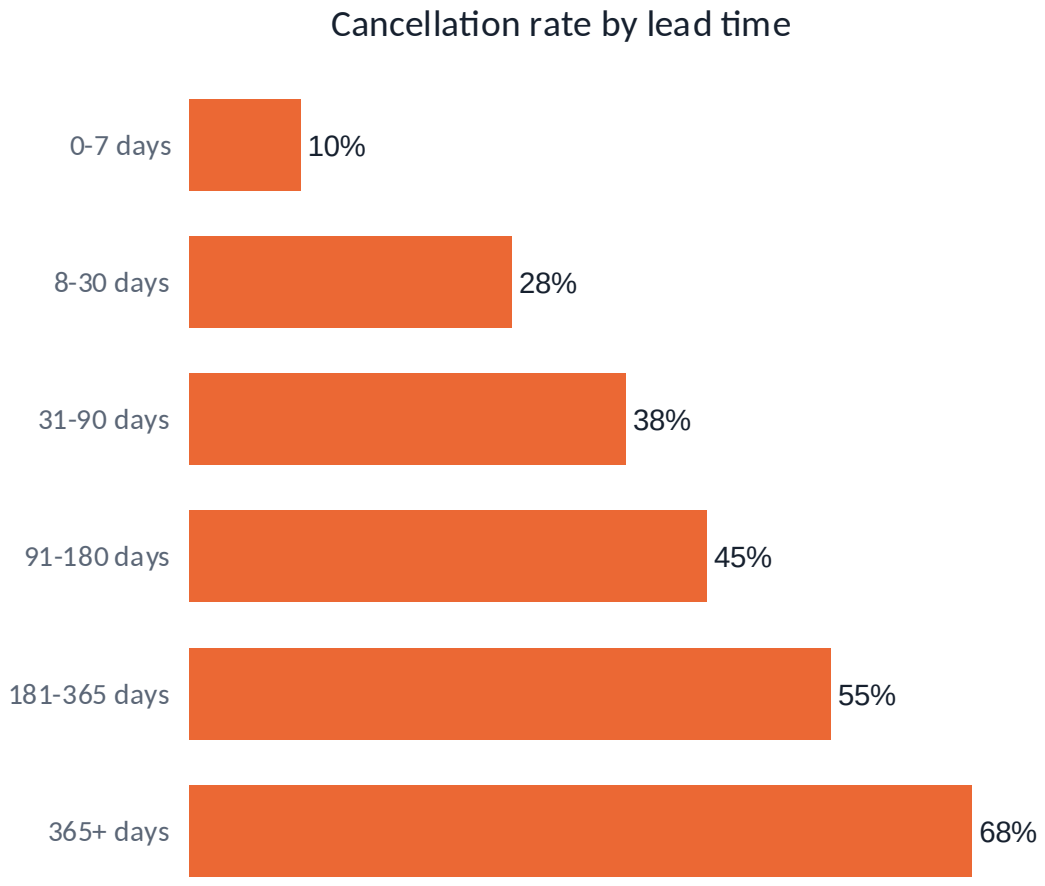
The City Hotel brings two-thirds of bookings but cancels far more; the Resort Hotel earns more per stay through longer summer holidays.

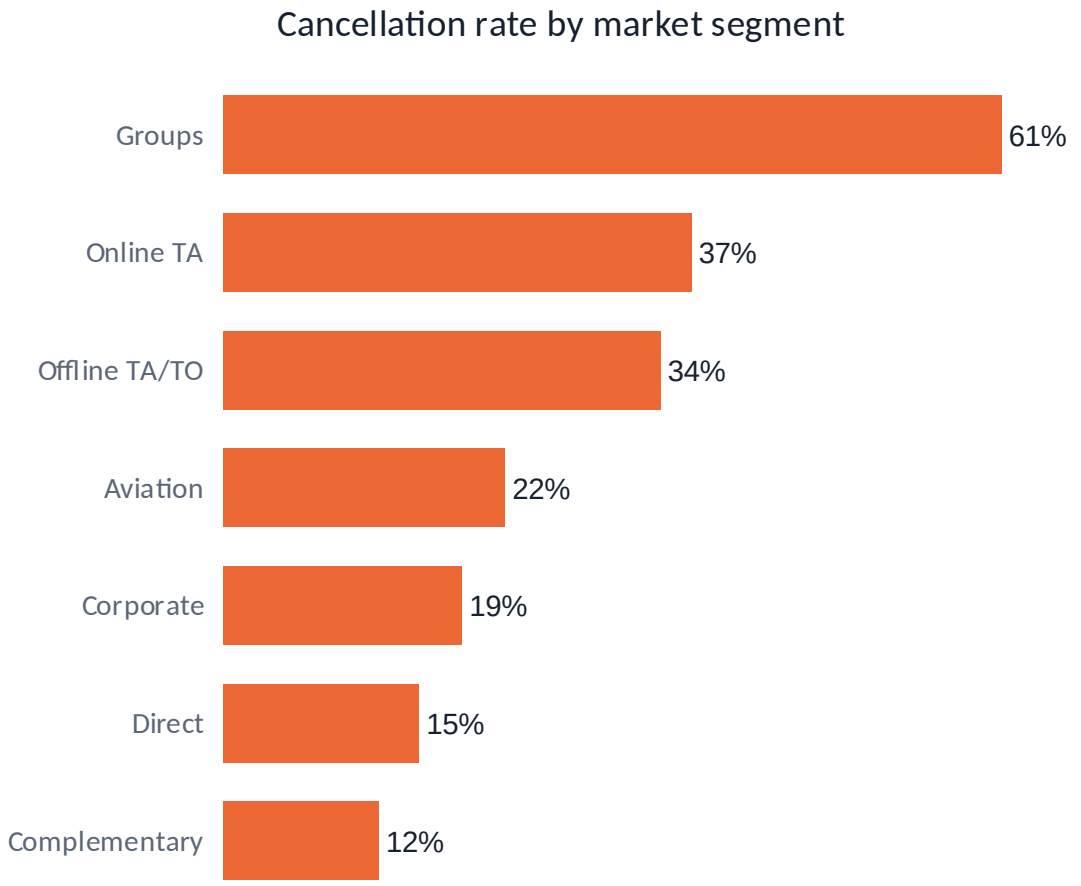
RO1 · BOOKINGS TREND

Both hotels grew, with a clear spring–autumn peak



Cancellations rise with lead time and group/OTA bookings





RO2 · CANCELLATION DRIVERS

Engaged guests stay; some signals need checking

0 SPECIAL REQUESTS

47.8%

vs 22.0% with 1 request

PARKING REQUESTED

0.0%

vs 39.5% without

REPEAT GUESTS

14.7%

vs 37.8% new guests

PREVIOUS CANCELLATION

91.7%

vs 33.9% without

PORTUGAL (DOMESTIC)

56.7%

vs 23.6% all other countries

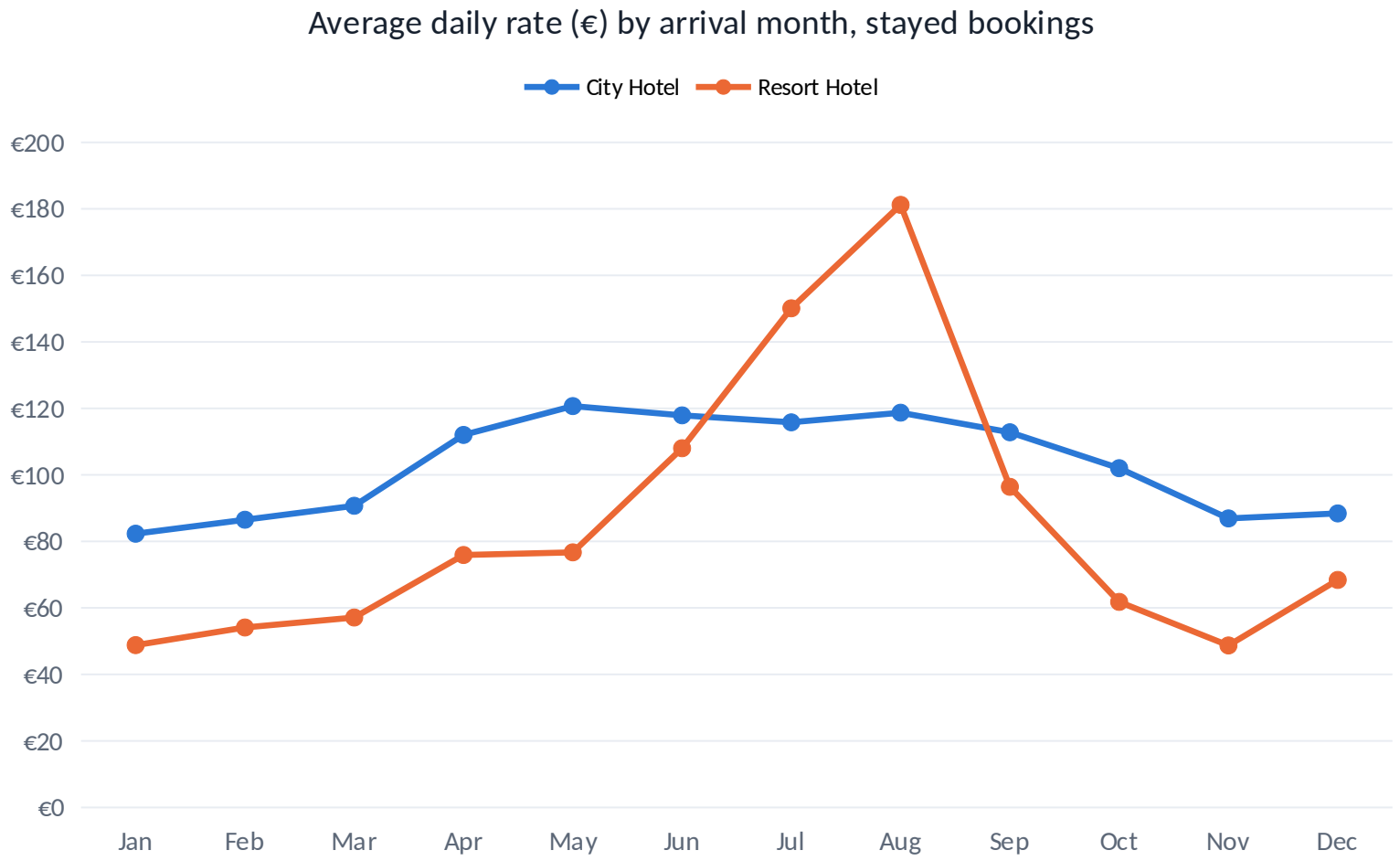
"NON REFUND" DEPOSIT

99.4%

data quirk – check with hotel

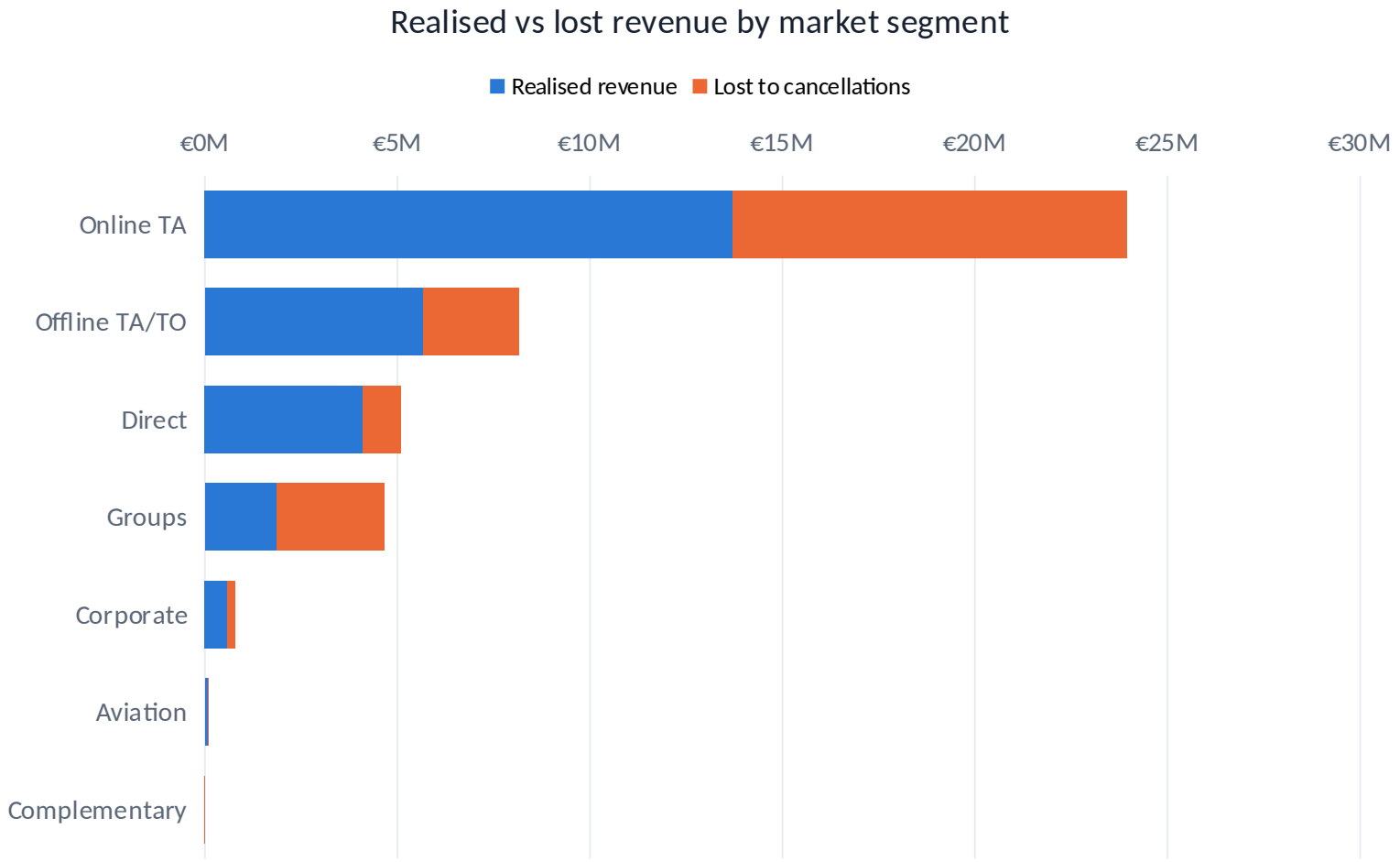
Non-refundable bookings should cancel least, not most. 97% are from Portugal and mostly Groups / Offline agents – likely held room blocks. Parking may be recorded at check-in. Both were flagged before using them for decisions.

Resort prices rise almost four-fold into August



- Resort: ~€49 in January → ~€181 in August
- City: steady €82 – €121 all year
- Summer = 45% of realised revenue
- Winter has the lowest cancellation rate (33%) – good for packages

Online agents earn most – and lose most

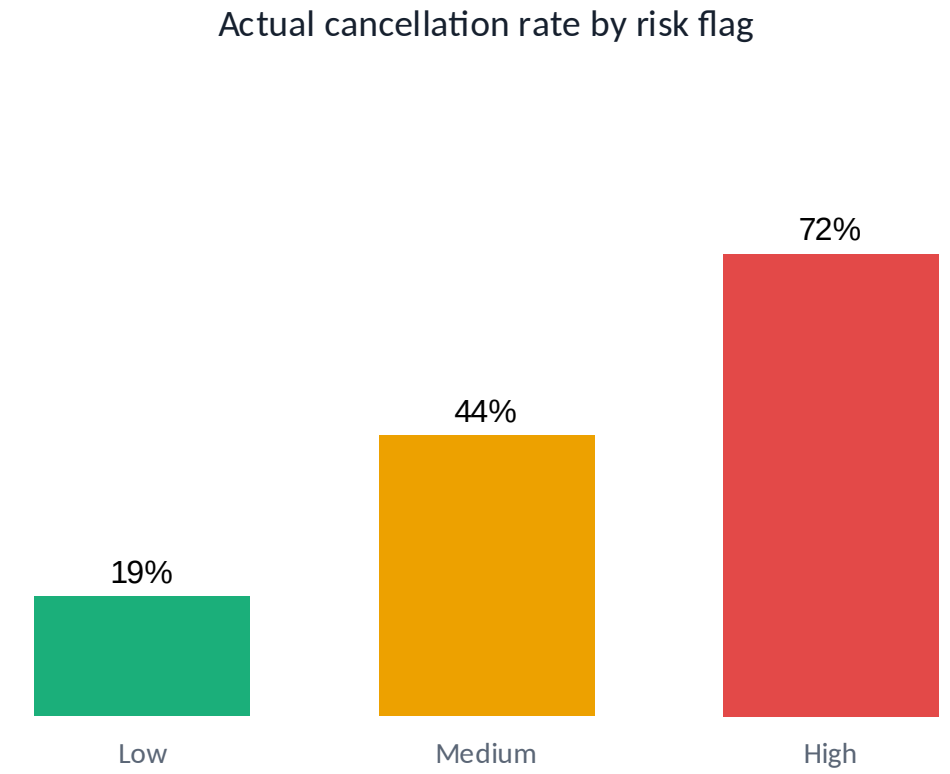


- Online TA: €13.7M earned, €10.2M lost
- Direct: same ADR as OTA (€114), 15% cancel, no commission
- Groups: lost €2.8M vs earned €1.9M
- → grow direct, tighten group & OTA terms

A simple risk flag: High-flag bookings cancel 72%

Risk factor (1 point each)	Rule
Long lead time	Over 90 days
High-cancel segment	Groups or Online TA
No special requests	0 requests
Flag	**0–1 Low · 2 Medium · 3 High**

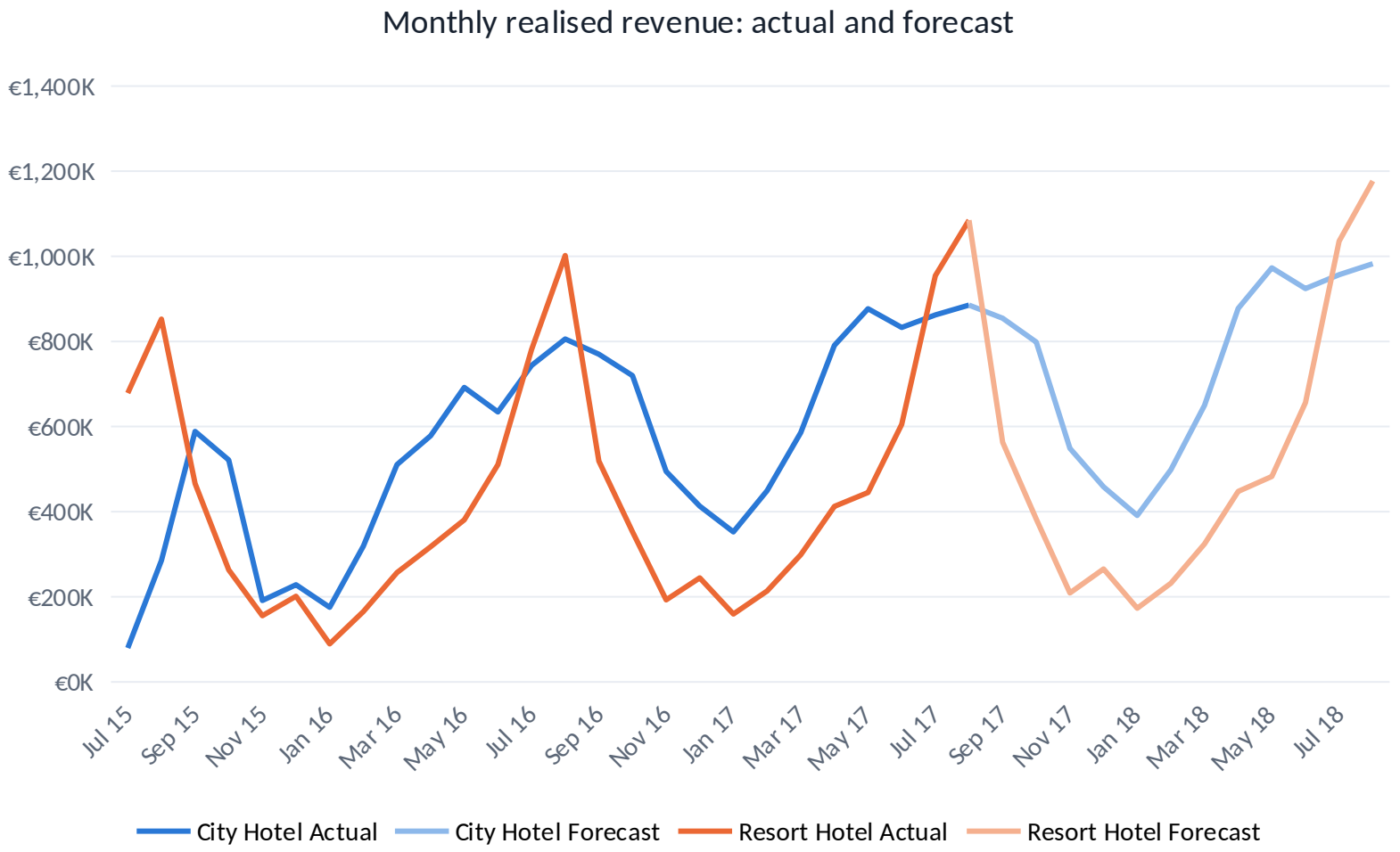
- Built from the three strongest findings (slides 7–8)
- Easy to explain, check and change – no special software
- Live formulas in Excel; calculated columns in Power BI
- Deposit type and parking left out (data-recording issues)



High flag: 20,427 bookings · €5.1M lost value · City Hotel High = 78%

RO5 · FORECAST

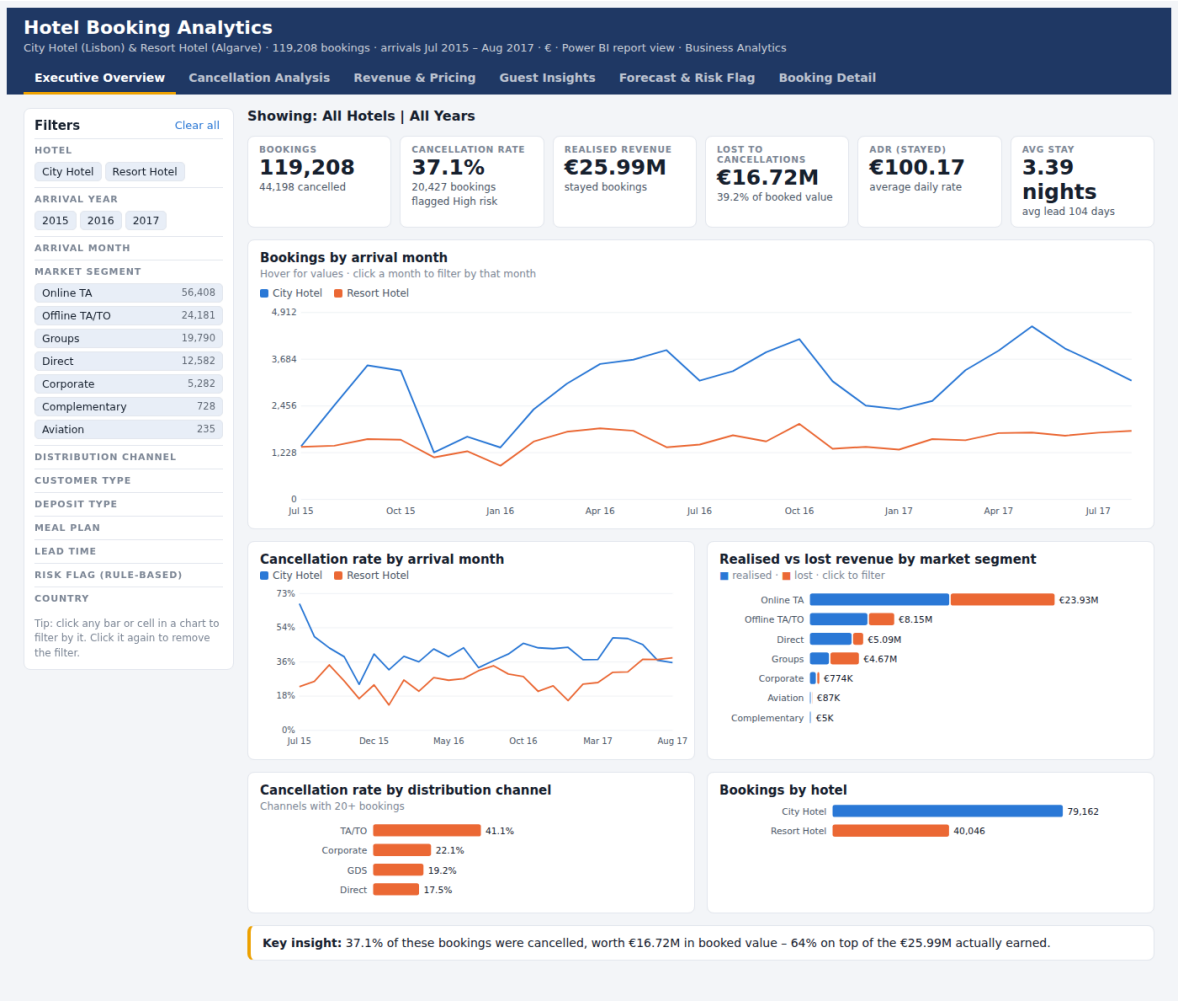
Next 12 months: revenue up ~10%, bookings up ~4%



- Method: same month last year × damped (50%) growth – runs in Excel
- Backtest error (MAPE): revenue 6–8%, bookings 11–12%
- City: €8.9M (+11.0%), 42.9K bookings (+4.4%)
- Resort: €5.9M (+8.5%), 19.7K bookings (+1.7%)

EXCEL + POWER BI

The business analytics system



Excel workbook

- Dashboard with Hotel & Year drop-downs (COUNTIFS / SUMIFS)
- 10 driver tables, forecast with editable damping, risk flag sheet
- 119K-row table ready for PivotTables & slicers

Power BI report – 6 pages

- Overview · Cancellations · Revenue & Pricing · Guests · Forecast & Risk Flag · Booking Detail
- Star schema, 20+ DAX measures, time intelligence, risk flag columns
- Slicers, drill-through, tooltips, Key Influencers, Q&A
- What-if slider: revenue recovered by cutting cancellations

- LINK TO DASHBOARD POWER BI
  LINK TO POWERBI

BUSINESS IMPACT

Recommendations

1. Deposit for High-flag bookings

Cut 20% of High-flag cancellations ≈ €1.0M



2. Smart overbooking

Count High/Medium flags per arrival date



3. Grow direct bookings

Same ADR, 15% vs 37% cancel, no commission



4. Tighter group & OTA terms

Groups lost €2.8M vs earned €1.9M



5. Engage long-lead guests

1+ request: 22% cancel vs 48%



6. Seasonal pricing & loyalty

Resort €49 → €181; repeat guests only 3%



BUSINESS IMPACT

Recommendations

1. Deposit for High-flag bookings

Cut 20% of High-flag cancellations ≈ €1.0M

€5.13M lost by High-flag bookings × 20% prevented



≈ €1.0M

value recovered

2. Smart overbooking

3. Grow direct bookings

4. Tighter group & OTA terms

5. Engage long-lead guests

6. Seasonal pricing & loyalty

RUBRIC ALIGNMENT

How the project meets the marking rubric

Marker	Evidence
Relevance & alignment (30%)	€16.7M problem; 5 objectives mapped to dashboard pages; decision table
Visualization & reporting (25%)	Consistent hotel colours, KPI cards, insight text on every page
Usability & engagement (20%)	Drop-downs, slicers, drill-through, tooltips, Q&A, what-if slider
Business impact (15%)	6 recommendations with € impact; risk flag on every booking
Data integration (5%)	5 tables → star schema; cleaning log; Power Query steps
Performance & scalability (5%)	Star schema & measures; incremental refresh; risk flag updates on refresh

Conclusion



THE PROBLEM

Cancellations – not demand – are the biggest revenue problem



THE PATTERN

They follow clear patterns: lead time, segment and guest engagement



THE ACTION

A simple risk flag shows which bookings to act on (High = 72% cancel)



THE OUTLOOK

Forecast: revenue up about 10% next year, mainly from price

Thank you